

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP8 enforced via flake8
2	Proper File Structure	Files and folders are logically organized	Yes	Separated into data/, model/, api/, and static/ modules
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. life_expectancy, hdi_score, sub_index_health
4	Function / Method Names	Functions are descriptively named	Yes	e.g. compute_hdi_score(), classify_tier()
5	Code Comments	Inline and block comments explain logic	Partial	Core model functions documented; UI helpers lightly commented
6	Modular Design	Code is split into reusable functions/modules	Yes	Sub-index calculation isolated from API and UI layers
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reviewed and cleaned before submission

8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation errors return clear API error messages
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	sub_index_calculator.py	Python	Model & API layers	High
2	hdi_classifier.py	Python (scikit-learn)	Prediction endpoint	High
3	ChartCard.jsx	React	Dashboard & Report pages	Medium
4	csv_batch_parser.py	Python (pandas)	Batch upload feature	Medium
5	report_generator.py	Python (reportlab)	PDF/CSV export feature	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean modular separation between data, model, and API layers.
Readability	4	Clear naming; some UI helper functions could use more comments.
Reusability	5	Core sub-index and report modules are reused across multiple endpoints.
Documentation / Comments	4	Model logic well documented; minor gaps in frontend comments.
Overall Score	4.5	Strong, maintainable codebase overall.